# Transducer

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# Introduction

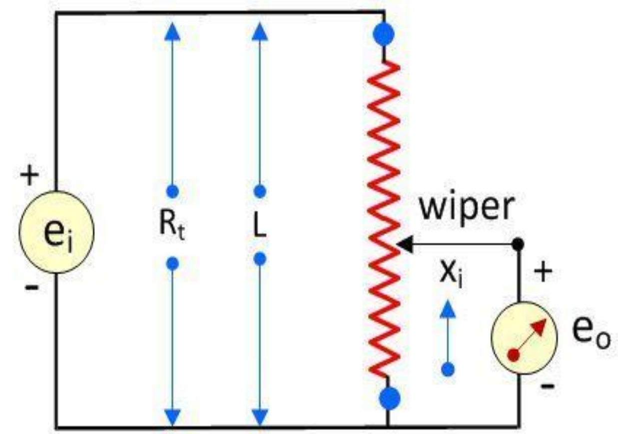
- Transducer is a electrical device which converts one form of energy into another form of energy.
- Transducer is a device which converts **physical or non-electrical parameter** (Temperature, pressure, displacement etc) into **electrical parameter**.
- Examples are microphones, Photovoltaic cell, resistance temperature detector, pressure sensor etc.
- **Advantages of Transducer-**
  - **Electrical amplification** and **attenuation** can be done easily with static device
  - The **mass-inertia** effects are **minimized**.
  - The **effects of friction** are **minimized**.
  - The electrical or electronic systems can be controlled with a very **small power level**.
  - The **electrical output** can be easily **transmitted** over a **long distance.**

# Classification of Transducer

- *Transducers are* ***classified*** *as-*
- ***Source of Energy-*** *Passive and Active Transducer.*
- ***Principle of Transduction.***
- ***Primary and Secondary Transducer.***
- ***Analog and Digital Transducer.***
- ***Transducer and Inverse Transducers.***

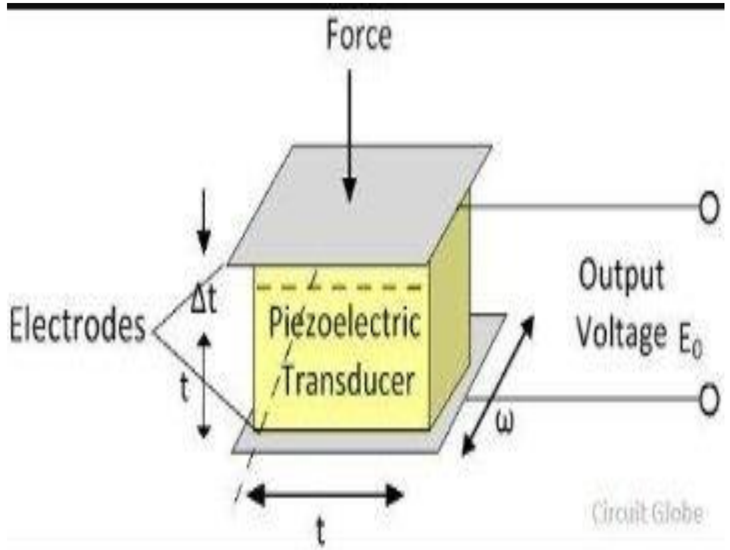
# Source of Energy

- **Passive Transducer-** Passive transducer derive the **required power** for transduction from the **auxiliary power source**. They are also known as **externally powered transducer**. Examples of passive transducers are **resistive, inductive and capacitive** transducer.



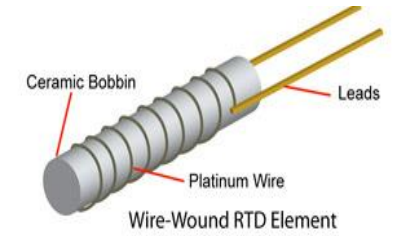

Linear Potentiometer (Pot), a passive transducer

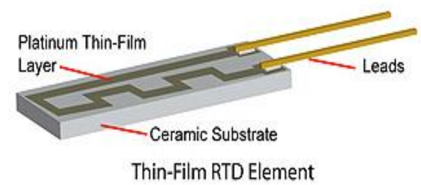
- **Active transducers-** **Active transducer** **does not require** an **auxiliary power source** to produce their output. They are also known as **self generating type transducer**. Examples are **thermocouple, photovoltaic, piezoelectric crystal** etc.

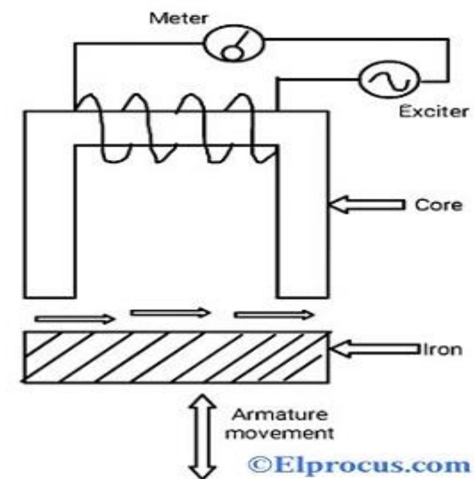


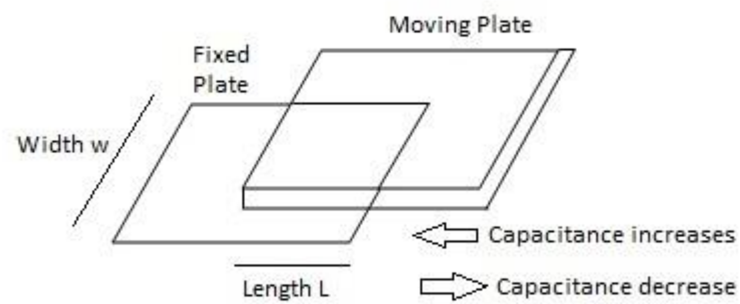
# Principle of Transduction

- The **transducers** can be classified on the basis of **principle of transduction** as **resistive, inductive, capacitive** etc.,
- If *resistance* varies with *change in physical parameter* then it is known as resistive type of transducer.
- If inductance varies with *change in physical parameter* then it is known as inductive type of transducer.
- If *capacitance* varies with *change in physical parameter* then it is known as capacitive type of transducer.



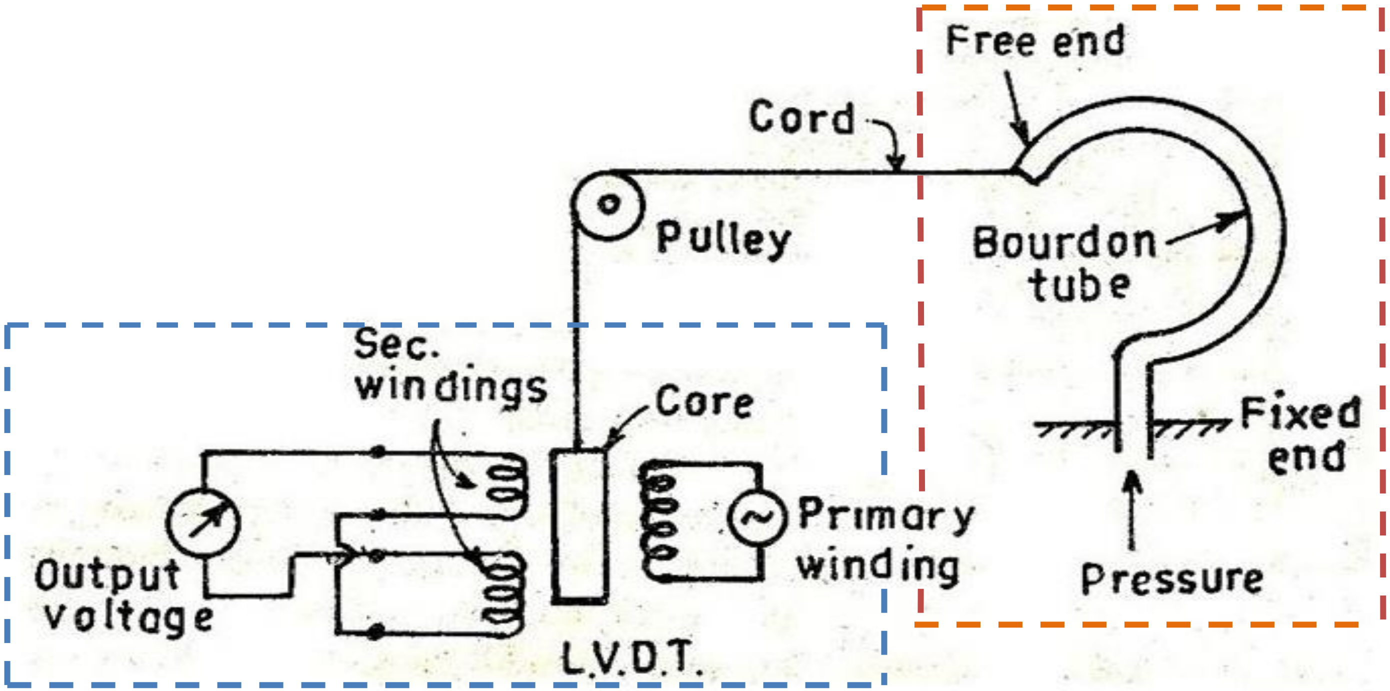



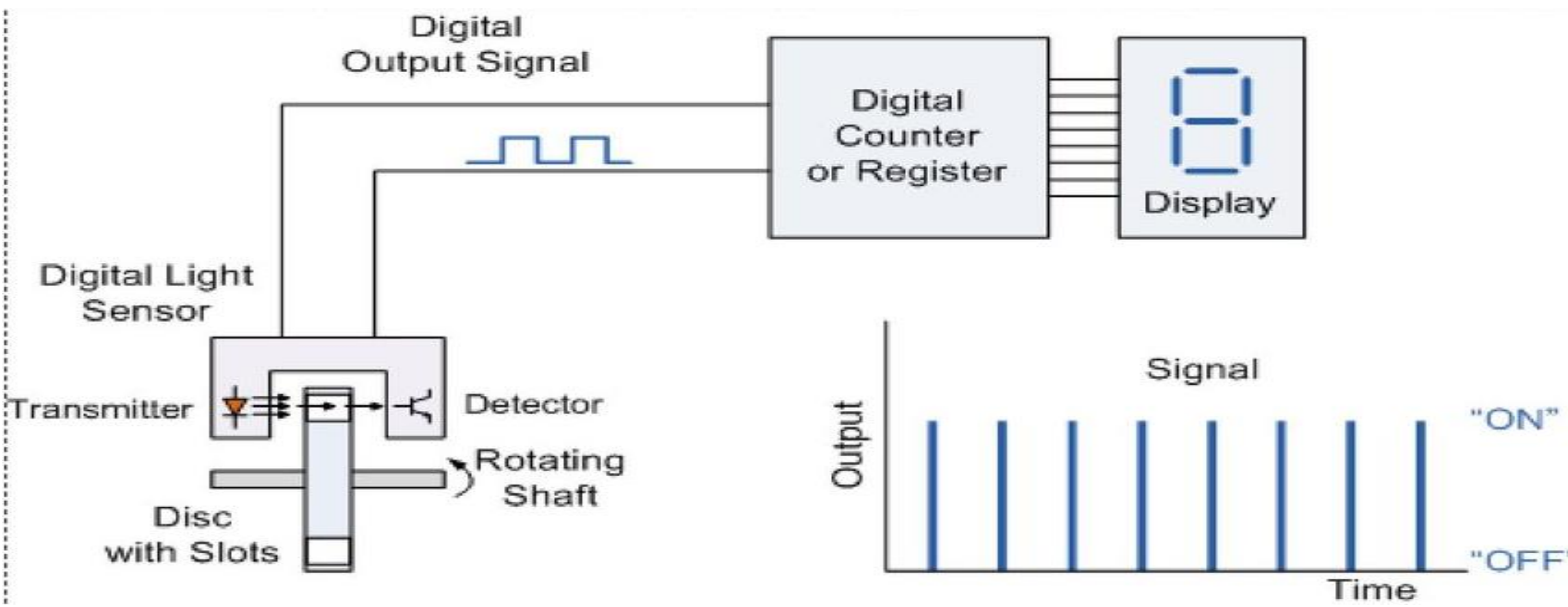
# Primary and Secondary Transducer

- The *transducer* which converts *physical quantity* into *mechanical quantity* is called *primary transducer*. It is a mechanical device.
- The transducer which converts *mechanical quantity* into *electrical output* is called *secondary transducer*. Output of primary transducer is converted into useful signal by secondary trans



# Analog and Digital Transducer

- **Analog Transducer** – The *Analog transducer changes* the *input quantity* into a *continuous function of time.* The *strain gauge, L.V.D.T, thermocouple, thermistor* are the examples of the analogue transducer.
- **Digital Transducer** – These *transducers convert* an input quantity into a *digital signal* or in the *form of the pulse*. The *digital transducer* works on *low power*.



# Transducer and Inverse Transducers

- Such instruments that is used to transform ***non-electrical parameters i***nto the ***electrical parameters*** are called a transducer.
- The device that is used to transform ***electrical parameters*** into the ***non-electrical parameters*** is called inverse transducers.